Back to course

Assignment

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100% (30/30)

Q1

6 / 6

Q1

Assigment

1

a)

i) Let $\vec{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{y} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

We have,

$$T_1(2\vec{x} + 0\vec{y}) = T_1\left(\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 4 \\ 2 \end{bmatrix} \neq 2T_1(\vec{x}) + 0T_1(\vec{y}) = 2T_1\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}$$

$\therefore$ T_1 is not a linear mapping by defn.

ii) Let $\vec{x}, \vec{y} \in \mathbb{R}^3$ $b, c \in \mathbb{R}$

We have,

$$\begin{aligned}
T_2(b\vec{x} + c\vec{y}) &= T_2(bx_1 + cy_1,\ bx_2 + cy_2,\ bx_3 + cy_3) \\
&= (2bx_2 + 2cy_2 + 3bx_3 + 3cy_3,\ 2bx_3 + 2cy_3 + 3bx_1 + 3cy_1,\ 2bx_1 + 2cy_1 + 3bx_2 + 3cy_2) \\
&= (b(2x_2 + 3x_3) + c(2y_2 + 3y_3),\ b(2x_3 + 3x_1) + c(2y_3 + 3y_1),\ b(2x_1 + 3x_2) + c(2y_1 + 3y_2)) \\
&= b(2x_2 + 3x_3,\ 2x_3 + 3x_1,\ 2x_1 + 3x_2) + c(2y_2 + 3y_3,\ 2y_3 + 3y_1,\ 2y_1 + 3y_2) \\
&= bT_2(\vec{x}) + cT_2(\vec{y})
\end{aligned}$$

So T_2 is a linear mapping by defn.

iii) $T_3(0,0,0) = (2(0) + 3,\ 5(0),\ 4(0) + 0)$

$= (3, 0, 0) \neq \vec{0}$

So T_3 is not a linear mapping by theorem.

b) Let $\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $\vec{v}_3 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

Notice $\vec{v}_3 = 2\vec{v}_1 - \vec{v}_2$

Assume there exists a linear mapping $T: \mathbb{R}^2 \to \mathbb{R}^3$ that satisfies the given condition.

Then by definition, $T(2\vec{v}_1 - \vec{v}_2) = 2T(\vec{v}_1) - T(\vec{v}_2)$

and $T(2\vec{v}_1 - \vec{v}_2) = T(\vec{v}_3)$ (since $\vec{v}_3 = 2\vec{v}_1 - \vec{v}_2$)

We have $T(2\vec{v}_1 - \vec{v}_2) = 2T(\vec{v}_1) - T(\vec{v}_2)$

$= 2(-2, 3, 4) - (1, 2, 3)$

$= (-5, 4, 5) \neq (1, -1, 1)$ $(= T(\vec{v}_3))$

$\therefore$ This is a contradiction to our assumption. Therefore, there does not exist such T.

Q2

4 / 4

Q2

(2) $T: \mathbb{R}^n \to \mathbb{R}^m$

a) ($\Rightarrow$) Assume T is onto

$\Rightarrow$ $\text{Range}(T) = \mathbb{R}^m$ (by defn) (1)

Since T is linear, we also have the standard matrix

$[T] = [T(\vec{e}_1) \cdots T(\vec{e}_n)]$

By theorem 3.3.27, since T is linear and has the standard matrix $[T]$,

$\text{Range}(T) = \text{Span}\{T(\vec{e}_1), \cdots, T(\vec{e}_n)\}$ (2)

From (1) and (2), we can conclude that $\text{Span}\{T(\vec{e}_1), \cdots, T(\vec{e}_n)\} = \mathbb{R}^m$.

($\Leftarrow$) Assume $\text{Span}\{T(\vec{e}_1), \ldots, T(\vec{e}_n)\} = \mathbb{R}^m$

Let $\vec{y} \in \mathbb{R}^m$, then $\exists\, c_1, \ldots, c_n \in \mathbb{R}$ such that

$$\vec{y} = c_1 T(\vec{e}_1) + c_2 T(\vec{e}_2) + \cdots + c_n T(\vec{e}_n)$$
$$= T(c_1\vec{e}_1 + c_2\vec{e}_2 + \ldots + c_n\vec{e}_n) \quad \text{(by linearity)}$$

Let $\vec{x} = c_1\vec{e}_1 + c_2\vec{e}_2 + \cdots + c_n\vec{e}_n \in \mathbb{R}^n \Rightarrow T(\vec{x}) = \vec{y}$

We have shown for every $\vec{y} \in \mathbb{R}^m$, $\exists\, \vec{x} \in \mathbb{R}^n$ such that $\vec{y} = T(\vec{x})$

$\therefore T$ is onto ▨

b) False

c) True

Q3

6 / 6

Q3

③ $L: \mathbb{R}^3 \to \mathbb{R}^3$ $\quad L(x_1, x_2, x_3) = (x_1 + x_3,\ x_1 + x_2 + 2x_3,\ 3x_1 + x_2 + 4x_3)$

$\vec{n} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ $\quad T: \mathbb{R}^3 \to \mathbb{R}^3 \quad T = L \circ \text{refl}_p$

a) . Standard matrix $[L]$:

$L(1,0,0) = (1,1,3) \qquad L(0,1,0) = (0,1,1) \qquad L(0,0,1) = (1,2,4)$

So $[L] = [L(\vec{e}_1)\ L(\vec{e}_2)\ L(\vec{e}_3)] = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 3 & 1 & 4 \end{bmatrix}$

. Standard matrix $[\text{refl}_p]$: $\quad \text{refl}_p \vec{x} = \vec{x} - 2\,\text{proj}_{\vec{n}} \vec{x} = \vec{x} - 2 \dfrac{\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \cdot \vec{x}}{2} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$

$\text{refl}_p \vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \qquad \text{refl}_p \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \qquad \text{refl}_p \vec{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} - 0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

So $[\text{refl}_p] = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

. Standard matrix $[T]$:

$[T] = [L \circ \text{refl}_p] = [L][\text{refl}_p] = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 3 & 1 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 3 & 4 \end{bmatrix}$

b) $\text{null}([T]) = \{\vec{x} \in \mathbb{R}^n \mid [T]\vec{x} = \vec{0}\}$

We want to solve:

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 3 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix} \begin{matrix} R_1 \Leftrightarrow R_2 \\ \\ R_3 - R_2 \end{matrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{matrix} R_1 - R_2 \\ \\ R_3 - 2R_2 \end{matrix}$$

$\Rightarrow \vec{x} = t \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, \ t \in \mathbb{R}$

$\Rightarrow \text{null}([T]) = \text{Span}\left\{\begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}\right\}$

$\Rightarrow \left\{\begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}\right\}$ is the basis for $\text{null}([T])$

$\therefore$ $\text{null}([T])$ is a 1-dimensional subspace (line) with the direction vector $\begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$ in $\mathbb{R}^3$.

c) $\text{Col}([T]) = \text{Span}\left\{\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}\right\} = \text{Span}\left\{\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}\right\}$ (Since $\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$)

. $\left\{\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}\right\}$ is linearly indep $\Rightarrow \left\{\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}\right\}$ is a basis for $\text{Col}([T])$

$\therefore$ $\text{Col}([T])$ is a 2-dimensional subspace (plane) in $\mathbb{R}^3$.

Q4

4 / 4

Q4

④ $L: \mathbb{R}^n \to \mathbb{R}^n$ $\quad$ Range(L) = Ker(L)

a) Let $\vec{x} \in \mathbb{R}^n$, and $\vec{y} = L(x)$

$\Rightarrow \vec{y} \in$ Range(L)

Since Range(L) = Ker(L), $\vec{y} \in$ Ker(L)

$\Rightarrow L(\vec{y}) = \vec{0}$ $\quad (*)$

Substitute $\vec{y} = L(\vec{x})$ into $(*)$ we have

$L(L(\vec{x})) = \vec{0}$

$\Rightarrow (L \circ L)(\vec{x}) = \vec{0}$ $\quad$ (by linearity)

$\Rightarrow [L \circ L] = 0_{n,n}$ $\quad$ (by linearity)

$\Rightarrow [L][L] = 0_{n,n}$

$\therefore [L]^2 = 0_{n,n}$ $\quad \blacksquare$

b) Disproof:

Counter example: Let $[L] = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

. Range(L) = Col$([L])$ = Span$\left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$ = Span$\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$

. $[L]\vec{x} = \vec{0}$, $\vec{x} \in \mathbb{R}^2$

$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $\Rightarrow x = t \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $(t \in \mathbb{R})$ $\Rightarrow$ Ker(L) = Span$\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$

So Range(L) = Ker(L), however $[L] = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \neq 0_{2,2}$

Q5

4 / 4

Q5

correct 4

(5) $tr(A) = \sum_{i=1}^{n} a_{ii}$

a) $A, B, C \in M_{n\times n}(\mathbb{R})$

$tr(AB) = \sum_{i=1}^{n} (AB)_{ii} = \sum_{i=1}^{n}\sum_{k=1}^{n} (A)_{ik}(B)_{ki}$ (1)

$tr(BA) = \sum_{i=1}^{n} (BA)_{ii} = \sum_{i=1}^{n}\sum_{k=1}^{n} (B)_{ik}(A)_{ki} = \sum_{i=1}^{n}\sum_{k=1}^{n} (A)_{ki}(B)_{ik}$ (2) (multiplication of real numbers is commutative)

Notice that (1) and (2) are both double sums over the same indicies i and k. Thus,

$tr(AB) = tr(BA)$

b) Disproof:

Q6

3 / 3

Q6

(6) $\|A\vec{x}\| = \|\vec{x}\|$ for all $\vec{x} \in \mathbb{R}^n$

a) We have $\|A\vec{x} - A\vec{y}\| = \|A(\vec{x} - \vec{y})\|$

Let $\vec{z} = \vec{x} - \vec{y}$. Since $\vec{x}, \vec{y} \in \mathbb{R}^n$, $\vec{x} - \vec{y} \in \mathbb{R}^n \Rightarrow \vec{z} \in \mathbb{R}^n$

Then $\|A\vec{z}\| = \|\vec{z}\|$

i.e $\|A(\vec{x} - \vec{y})\| = \|\vec{x} - \vec{y}\|$

$\therefore \|A\vec{x} - A\vec{y}\| = \|\vec{x} - \vec{y}\|$ ▨

b) Let $\vec{x}, \vec{y} \in \mathbb{R}^n$

From part a, we have

$$\|A\vec{x} - A\vec{y}\| = \|\vec{x} - \vec{y}\|$$

$$\|A\vec{x} - A\vec{y}\|^2 = \|\vec{x} - \vec{y}\|^2$$

$$(A\vec{x} - A\vec{y})\cdot(A\vec{x} - A\vec{y}) = (\vec{x} - \vec{y})(\vec{x} - \vec{y})$$

$$A\vec{x}\cdot A\vec{x} - 2A\vec{x}\cdot A\vec{y} + A\vec{y}\cdot A\vec{y} = \vec{x}\cdot\vec{x} - 2\vec{x}\cdot\vec{y} + \vec{y}\cdot\vec{y}$$

$$\|A\vec{x}\|^2 - 2A\vec{x}\cdot A\vec{y} + \|A\vec{y}\|^2 = \|\vec{x}\|^2 - 2\vec{x}\cdot\vec{y} + \|\vec{y}\|^2$$

$$\|\vec{x}\|^2 - 2A\vec{x}\cdot A\vec{y} + \|\vec{y}\|^2 = \|\vec{x}\|^2 - 2\vec{x}\cdot\vec{y} + \|\vec{y}\|^2$$

$$A\vec{x}\cdot A\vec{y} = \vec{x}\cdot\vec{y}$$ ▨

Q7

3 / 3

Q7

(7) Proof:

Assume for the sake of contradiction that $AB - BA = I$ for some $A, B \in M_{n\times n}(\mathbb{R})$

Take the trace of both sides we have $tr(AB - BA) = tr(I)$

$tr(AB - BA) = \sum_{i=1}^{n} (AB + BA)_{ii} = \sum_{i=1}^{n} [(AB)_{ii} + (BA)_{ii}] = \sum_{i=1}^{n} (AB)_{ii} + \sum_{i=1}^{n} (BA)_{ii} = tr(AB) - tr(BA)$

$= 0$ (as proven in Q5)

Good

$\Rightarrow tr(I) = tr(AB - BA) = 0$

$\Rightarrow \sum_{i=1}^{n} I_{ii} = 0$

We know that I_{ii} in the identity matrix in $\mathbb{R}^n$ will always equals 1

$\Rightarrow \sum_{i=1}^{n} I_{ii} = n = 0$

This is a contradiction since it is not possible for matrices of size $n \times n$ (n must ≥ 1) $\boxtimes$